Climate Resilience: A Multi-Source Analysis of Climate Policy Evolution Post Paris Agreement (2013–2025)

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This application report documents a data mining pipeline which brings together three separate sources of climate data: the Climate Policy Database (CPDB), the UNFCCC NDCs and the Notre Dame Global Adaptation Initiative (ND-GAIN).The exploration analyzes the evolution of climate strategies since the Paris agreement(2015), with a particular focus on the integration of adaptation measures alongside mitigation efforts.

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1 Introduction

Climate policy research traditionally focuses on mitigation, with an emphasis on the reduction of greenhouse gas emissions. However, the increasing frequency and severity of climate impacts has elevated the importance of adaptation measures in national policy portfolios. The Paris Agreement (2015) marked a pivotal moment in global climate governance, with its dual focus on mitigation and adaptation.

As the Grantham Research Institute notes, "unlike climate mitigation, where responses like carbon pricing and emission targets are widely tracked, adaptation laws and policies have received less systematic scrutiny" [1, p.10]. International climate databases systematically under represent adaptation policies, particularly in the Global South where adaptation is a **survival imperative** rather than a discretionary policy choice. When adaptation is invisible in the data, it becomes invisible in policy debates, climate finance allocation and accountability mechanisms. The lack of "comparative global analysis and systematic scrutiny of national adaptation legislation and policies has hampered the ability to assess progress, identify gaps, and understand effective frameworks" [1, p.10]. This project serves as a digital audit of climate ambition across our selection of countries.

Comparing the Global North to the Global South is a major focus of this project. The Global North has historically been responsible for the majority of emissions and has more resources to invest in both mitigation and adaptation. The Global South, whilst being less responsible for the acceleration of climate change, are the ones who bear the brunt of it.

2 Research Question

Have climate policy portfolios shifted from mitigation toward adaptation after the Paris Agreement, and to what extent does this shift differ between Global North and Global South countries?

Key Terms:

- **Climate Policy Portfolio**: The collection of all climate-related policies enacted by a country.
- **Mitigation**: Efforts to reduce or prevent the emission of greenhouse gases.
- **Adaptation**: Efforts to adjust to actual or expected climate change and its effects

Technical Decisions:

The Temporal Boundary (2013–2025)

The choice of 2013 as the starting point is strategic. It allows us to capture the policy landscape leading up to the Paris Agreement, providing a baseline for comparison. The end point of 2025 ensures we have the most recent data available.

Country Selection

The selection of 12 countries is designed to provide a diverse sample that includes both Global North and Global South nations.

The logic is as follows:

Country	ISO-3 Code	Region	ND-GAIN
Canada	CAN	North	Low
Germany	DEU	North	Low
Japan	JPN	North	Low
United States	USA	North	Low
United Kingdom	GBR	North	Low
Denmark	DNK	North	Low
Saudi Arabia	SAU	South	Medium
Chile	CHL	South	Medium
India	IND	South	High
South Africa	ZAF	South	High
Colombia	COL	South	High
Rwanda	RWA	South	High

3 The Data Ecosystem

This project triangulates three distinct data sources:

3.1 NewClimate Institute's Climate Policy Database (CPDB)

The CPDB provides the quantitative baseline for actual policy adoption. Accessed via a Python-R bridge (reticulate), it offers broader geographic coverage than the OECD dataset that was considered (OECD CAPMF), including key Global South nations. Its primary limitation is a "Global North bias," where sub national or informal adaptation measures in developing regions are often under-indexed. This is a gap addressed by the inclusion of NDC texts.

3.2 UNFCCC Nationally Determined Contributions (NDCs)

To capture strategic intent beyond formal legislation, I developed an automated pipeline to extract and analyze the most recent NDC PDFs using the pdftools package. These documents serve as primary statements of government priority. While they allow for a nuanced NLP analysis of "Lexical Hardness" (Hard vs. Soft law), they represent diplomatic pledges rather than tangible enforcement.

3.3 ND-GAIN Index (Vulnerability Benchmarking)

The ND-GAIN Index provides the context of necessity. By integrating composite vulnerability and readiness scores, I was able to classify the 12 target countries by their actual exposure to climate hazards.

4 The Technical Pipeline

The entire workflow is orchestrated by a single master script (`main.R`) that manages:

1. **Dependency installation**
2. **Data acquisition**
3. **Data processing** (cleaning, merging, feature engineering).
4. **Analysis**
5. **Output generation** (figures, tables, this report).
6. **Environment Cleaning** (removing temporary files and data).

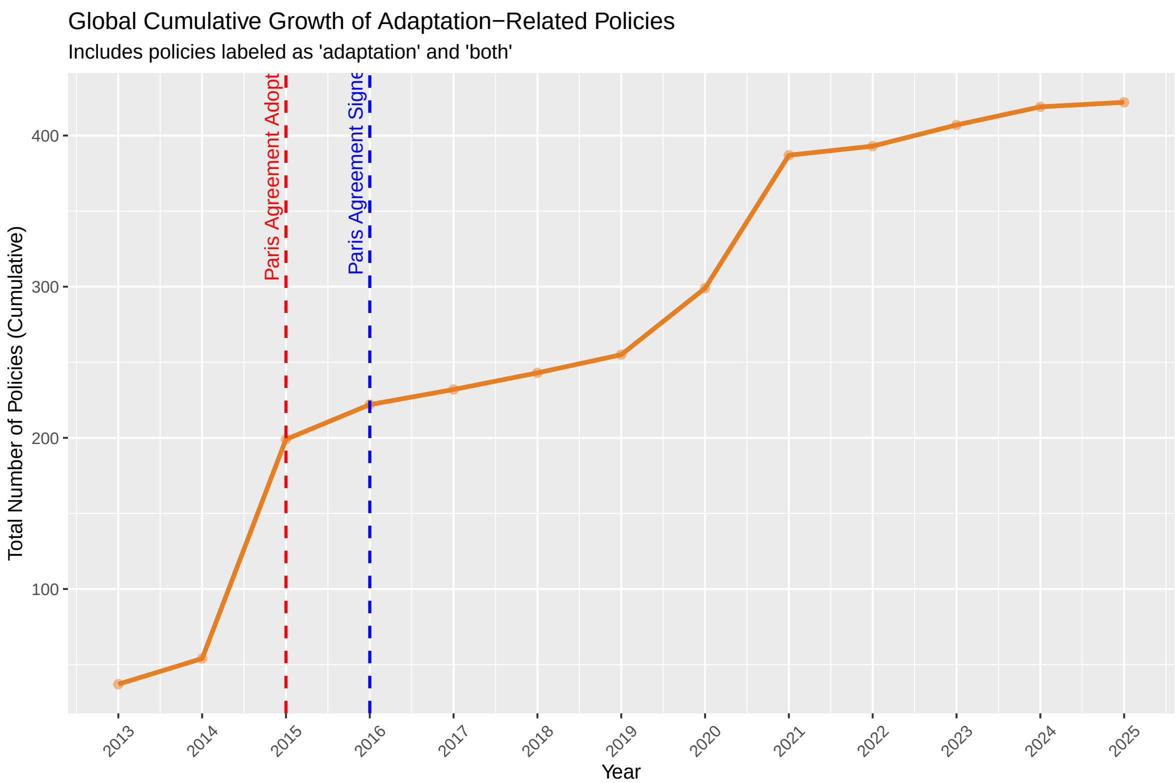


Figure 1: Global Cumulative Growth of Adaptation Policies (2013-2025)

5 Results

5.1 Global Adaptation Growth

Initial analysis of climate policies in the cpdb reveals a visible acceleration in global adaptation policy adoption following the 2015 Paris Agreement. This is the starting point for our analysis, confirming that the Paris Agreement did indeed catalyze a shift in policy focus.

5.2 Policy Implementation per Country (The Adaptation Mix)

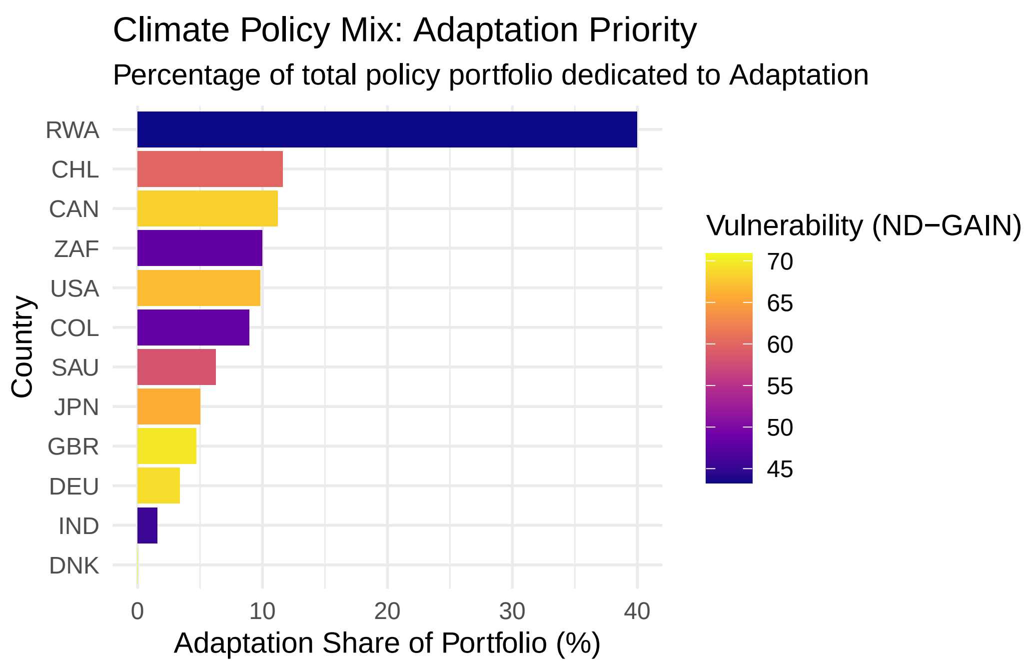
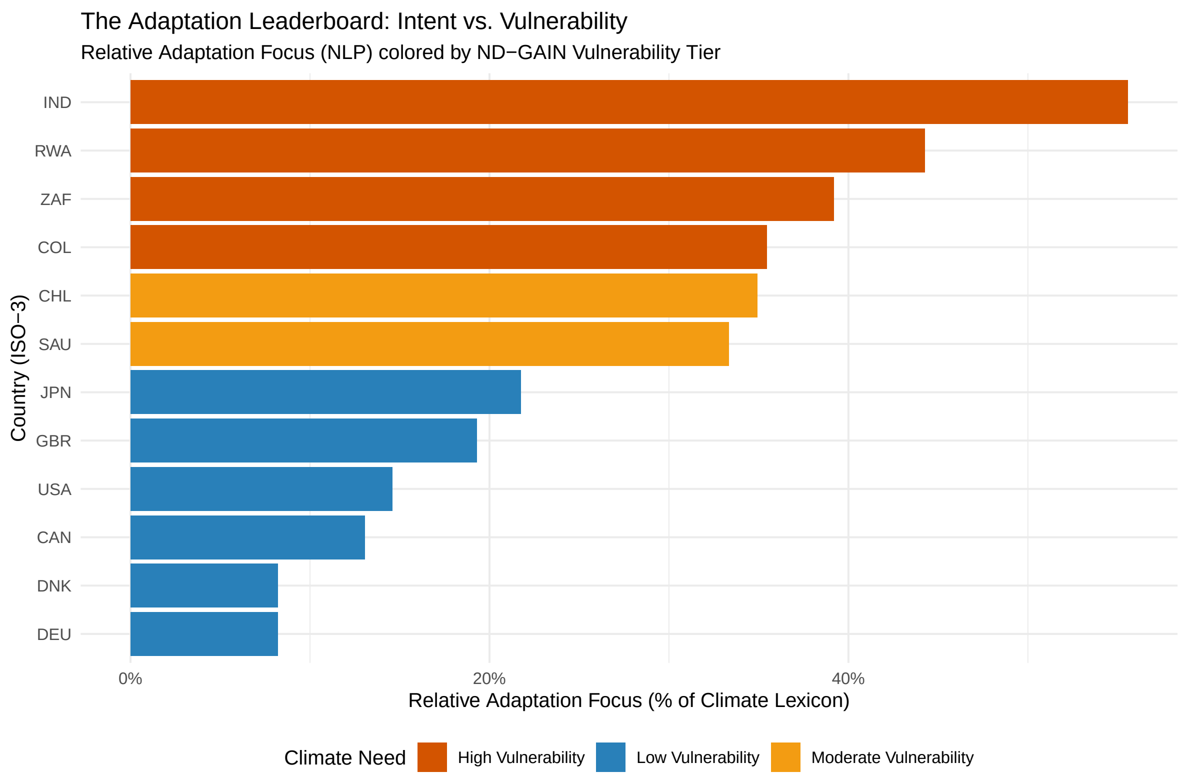


Figure 2: The Adaptation Mix: Percentage of Portfolio focusing on Adaptation

The "Adaptation Mix" (Figure 3) illustrates an uneven relationship between vulnerability and implementation. While some vulnerable nations like Rwanda show a high adaptation ratio, others like India lag behind.This would suggest that vulnerability may not be a driving factor in adaptation adoption, however this dataset is largely biased towards Global North countries, so it may not be capturing the true adaptation efforts of the Global South. This is a key limitation of the CPDB data and highlights the need to look at other sources (like the NDCs) to get a fuller picture of adaptation focus, particularly in the Global South where adaptation is more critical.

5.3 Adaptation Focus in NDCs (The Responsibility Gap)



(a) Adaptation Focus Score (Keyword Frequency) in NDCs, Sorted by ND-GAIN Vulnerability

Figure 3: Adaptation Focus in NDCs: Global South vs. Global North

By shifting to an NLP-based analysis of NDCs (Figure 4), the correlation between physical risk and strategic intent becomes strikingly clear. High-vulnerability countries dedicate significantly more "mindshare" to adaptation in their NDCs compared to the Global North. This is the Responsibility Gap: the nations least responsible for historical emissions are the ones forced to prioritize survival strategies.

The Ambition Audit			
Comparing Policy Implementation (CPDB) to Strategic Focus (NDC)			
ISO-3	CPDB Mix (%)	NDC Focus (%)	The Gap (diff)
IND	1.6	55.6	54.0
RWA	40.0	44.3	4.3
ZAF	10.0	39.2	29.2
COL	8.9	35.4	26.5
CHL	11.6	34.9	23.3
SAU	6.2	33.3	27.1
JPN	5.0	21.7	16.7
GBR	4.7	19.3	14.6
USA	9.8	14.6	4.8
CAN	11.2	13.0	1.8
DEU	3.4	8.2	4.8
DNK	0.0	8.2	8.2

Figure 4: Table 1: The Ambition Audit

Table 1 shows a comparison between the CPDB adaptation mix and the NDC adaptation focus. The gap between intent and policy implementation is particularly wide for the Global South, where adaptation is a critical need but resource constraints limit actual policy adoption. This brings about two important points 1) the CPDB data is likely underrepresenting the true adaptation efforts of the Global South, and 2) there is a significant gap between strategic intent (NDCs) and actual policy implementation (CPDB), particularly for adaptation in the Global South. This is the Commitment Gap: vulnerable countries are prioritizing adaptation in their strategic documents but are unable to translate this into binding policies due to resource limitations. A further understanding of this commitment gap is provided by figure 6, which maps the relative adaptation focus against the lexical hardness of the NDC language.

5.4 Adaptation Focus vs. Ambition in NDCs (The Commitment Gap)

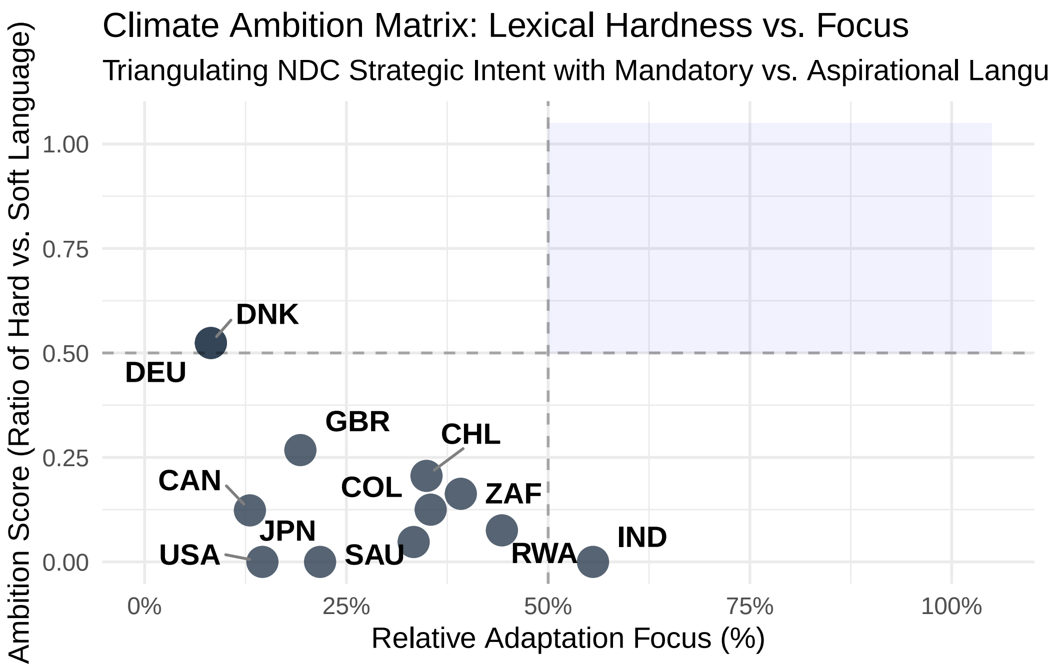


Figure 5: Climate Ambition Matrix: Lexical Hardness vs. Adaptation Focus

The Ambition Matrix (Figure 6) maps the final dimension of the Commitment Gap. The Ambition Score is calculated based on the frequency of "hard" vs. "soft" language in the NDC texts, with "hard" language indicating more binding commitments. Global North countries cluster in the "low focus, high hardness" quadrant, backed by legally binding mitigation mandates. Conversely, the Global South occupies the "high focus, low hardness" zone. This reflects structural inequalities. Vulnerable countries cannot legally bind themselves to adaptation measures without guaranteed external finance.

6 Limitations

A primary limitation of this project is the reliance on the CPDB for policy data, which is not exhaustive and has a bias toward Global North countries. This means that the adaptation policies of many Global South countries may be underrepresented, potentially skewing the analysis.

The NDC analysis also has limitations due to the variability in language and the challenge of quantifying qualitative commitments. The execution of the Lexical Audit was constrained by the depth of the text-mining heuristics. The current pipeline relies on a keyword-based dictionary approach which may fail to capture complex nuances. Future iterations would benefit from a more robust NLP framework to move beyond simple frequency counts and more accurately distinguish between aspirational rhetoric and enforceable domestic mandates.

7 Discussion

The four key findings provide empirical evidence for climate justice debates: the Global South prioritizes adaptation, but lacks the resources to codify commitments into binding law.

The Responsibility and Commitment Gaps both provide important evidence that it is those who are most vulnerable to climate impacts (the Global South) who are thinking about adaptation the most and those who contributed the most to climate change (the Global North) who are thinking about it the least. This is a stark illustration of the injustice at the heart of climate change: those who are least responsible for causing it are the ones who suffer its worst effects and have the greatest need for adaptation measures, yet they have the least capacity to do so.

The Commitment Gap finding has particular policy significance. It demonstrates that Global South adaptation priorities are substantive (evidenced by high keyword frequency and development plans explicitly linking adaptation to poverty reduction [1, p.14-15]), but these priorities are constrained by resource limitations (manifested as soft law language because commitments are conditional on international finance [1, p.14]).

8 References

[1] Grantham Research Institute on Climate Change and the Environment, "Climate change adaptation laws and policies: Global trends and insights 2025," London School of Economics and Political Science, 2025.

9 LLM Declaration

The following LLMs have been used in the creating of this report and project:

- Gemini for technical troubleshooting, code review and idea exploration. I used Gemini throughout the coding process and also as a way to talk through my ideas. All code has been a joint effort, never a complete copy-paste but more of a back and forth discussion about best methods. I have a background in software development but have never used R before, so Gemini has helped me navigate the syntax.

- Github Copilot for code suggestions and autocompletion. I used Copilot in my R scripts to speed up coding and get suggestions for functions and methods. Again, all code has been a joint effort, with me writing the logic and Copilot suggesting syntax and functions. In some cases I have used comments to explain what I want to do and then Copilot has generated the code to accomplish that task, which I have then reviewed and edited as needed.
- Scispace for help writing this report. I used Scispace to help me structure this report. I have used Scispace to help organize my thoughts and ensure that I am covering all the necessary sections and points.

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